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CASE REPORT

Disseminated Cryptococcosis in a Non-HIV Patient: Diagnostic Value of Integrating mNGS, Culture, and Antigen Testing

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Abstract: *Cryptococcus neoformans* is an opportunistic fungal pathogen most commonly observed in individuals with human immunodeficiency virus (HIV) infection. Disseminated infections involving multiple organs, such as lungs, bloodstream, urinary tract, pleural cavity and central nervous system, are uncommon in patients without HIV. This work describes the case of a 72-year-old man who developed high fever and increased inflammatory markers after coronary artery bypass surgery. *C. neoformans* (8 sequence reads) was initially detected by metagenomic next-generation sequencing (mNGS) of bronchoalveolar lavage fluid and was subsequently isolated from blood, urine and pleural effusion culture. Additionally, the cerebrospinal fluid tested positive for cryptococcal capsular polysaccharide antigen (CrAg), confirming disseminated infection involving multiple anatomical sites. The patient received liposomal amphotericin B combined with flucytosine antifungal treatment, which led to improvement in inflammatory parameters; however, the patient developed secondary multiorgan failure due to the severity of the illness. After 98 days of hospitalization, the patient was eventually discharged. This case highlights the diagnostic challenges of disseminated cryptococcosis in non-HIV hosts, particularly when involving atypical sites such as the urinary tract and pleural cavity. The integration of conventional microbiological methods, CrAg testing, and mNGS facilitated early diagnosis and enabled timely, standardized antifungal therapy.

Keywords: disseminated cryptococcosis, non-human immunodeficiency virus, cryptococcal capsular polysaccharide antigen, metagenomic next-generation sequencing

Introduction

Cryptococcus neoformans is widely found in pigeon droppings, soil, and decaying wood. After inhalation, pulmonary infection starts in humans and hematogenous dissemination to the central nervous system (CNS) may subsequently occur. CNS involvement is most commonly manifested as cryptococcal meningitis (CM), a condition associated with high mortality.[1] Globally, CM is predominantly linked to human immunodeficiency virus (HIV) infection.[2] However, with the increasing application of organ transplantation, corticosteroids, immunosuppressive agents, and broad-spectrum antibiotics, a rising incidence of cryptococcal disease in non-HIV populations has been observed.[3,4] In China, 60–70% of cryptococcosis cases have been reported in individuals without HIV infection.[5]

Despite this epidemiological shift, diagnosing disseminated cryptococcosis in non-HIV patients remains a formidable clinical challenge. Among these patients, the absence of typical immunocompromising conditions and the presence of insidious nonspecific symptoms often result in delayed or missed diagnosis during initial evaluation, leading to markedly higher fatality rates than those seen in HIV-positive individuals.[6] This diagnostic dilemma is compounded by the limitations

https://doi.org/10.2147/IDR.S597466
Received: 11 February 2026
Accepted: 4 June 2026
Published: 16 June 2026

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inherent to any single test. Although conventional culture remains the gold standard for diagnosis, it is time-consuming and associated with a low positivity rate.[7] Cryptococcal antigen (CrAg) testing may yield false-negative results and exhibit cross-reactivity.[8,9] Metagenomic next-generation sequencing (mNGS), despite its high sensitivity, can generate low read counts that pose interpretative challenges.[10] Therefore, a critical knowledge gap exists regarding how to integrate multi-modal diagnostic techniques (mNGS, culture, and CrAg) to achieve early and accurate diagnosis of disseminated cryptococcosis in non-HIV patients, particularly when atypical sites such as the urinary tract and pleura are involved.

To address this gap, we describe a case of disseminated *C. neoformans* infection in a non-HIV patient following coronary artery bypass surgery, with disease confirmed by positive cultures from blood, urine, and pleural effusion, positive CSF CrAg, and detection of *C. neoformans* by mNGS of bronchoalveolar lavage fluid.

Case Presentation

A 72-year-old man experienced unexplained dull, self-resolving chest pain on March 6. As the pain worsened, medical attention was sought at a local hospital on March 8. Coronary angiography was performed, and a stent was placed in the mid-to-distal segment of the left anterior descending artery. The follow-up echocardiogram on March 14 demonstrated an anterior wall myocardial infarction involving the mid-interventricular septum and apex, accompanied by an apical ventricular aneurysm and a suspected ventricular septal rupture. His past medical history included diabetes mellitus, routinely managed with metformin.

Admission Examination

The patient was transferred to our hospital on March 17. On admission, body temperature was 36.7°C, heart rate 101 beats/min, respiratory rate 20 breaths/min, and blood pressure 84/50 mmHg. Laboratory testing showed a white blood cell count of 10.47×10^9/L, red blood cell count of 5.03×10^{12}/L, hemoglobin 129 g/L, platelet count 380×10^9/L, C-reactive protein 40.46 mg/L, alanine aminotransferase 53 U/L, aspartate aminotransferase 35 U/L, albumin 70 g/L, creatinine 231 μmol/L, and blood urea nitrogen 24.7 mmol/L. HIV testing was negative.

Microbiological Analysis and Clinical Course

On April 11 (Day 25 from the admission), coronary artery bypass grafting, ventricular septal defect repair, and ventricular aneurysmectomy were performed.

On May 12 (Day 57), the patient developed a high fever (peak 39.1°C) and increased inflammatory markers, with white blood cell count of 17.61×10^9/L, neutrophils 82.8%, and C-reactive protein 85.97 mg/L. Chest CT showed bilateral pulmonary edema with pleural effusion.

On May 19 (Day 63), infection progression was observed, with a white blood cell count of 28.47×10^9/L, neutrophils 25.05×10^9/L, procalcitonin 3.24 ng/mL, and C-reactive protein 64.73 mg/L. A follow-up chest CT revealed worsening right lung pneumonia.

On May 21 (Day 66), bronchoalveolar lavage fluid (BALF) was submitted for metagenomic next-generation sequencing (mNGS), which detected *C. neoformans* with 8 sequence reads and *Candida auris* with 2,649 sequence reads (both at 99% confidence; see Supplementary Table 1). However, routine fungal culture of the same BALF specimen was negative for both organisms.

On May 26 (Day 71), a blood culture was collected and submitted.

On May 31 (Day 76), the aerobic blood culture bottle resulted positive after 112.1 hours of incubation, and *C. neoformans* was identified by mass spectrometry (Figure 1).

On June 3 (Day 79), ultrasound examination indicated increased pleural effusion and turbid urine. Urine and pleural effusion cultures were submitted for analysis, and *C. neoformans* was isolated from both specimens. Antifungal susceptibility results are provided in Table 1.

On June 5 (Day 81), the patient experienced impaired consciousness. Lumbar puncture revealed an opening cerebrospinal fluid (CSF) pressure > 300 mmH_2O, and CSF cryptococcal antigen testing was positive. Although both CSF mNGS and culture were negative for *C. neoformans*, CSF biochemical analysis showed elevated protein (106 g/L), glucose 16.2 mmol/L, and chloride (112 mmol/L).

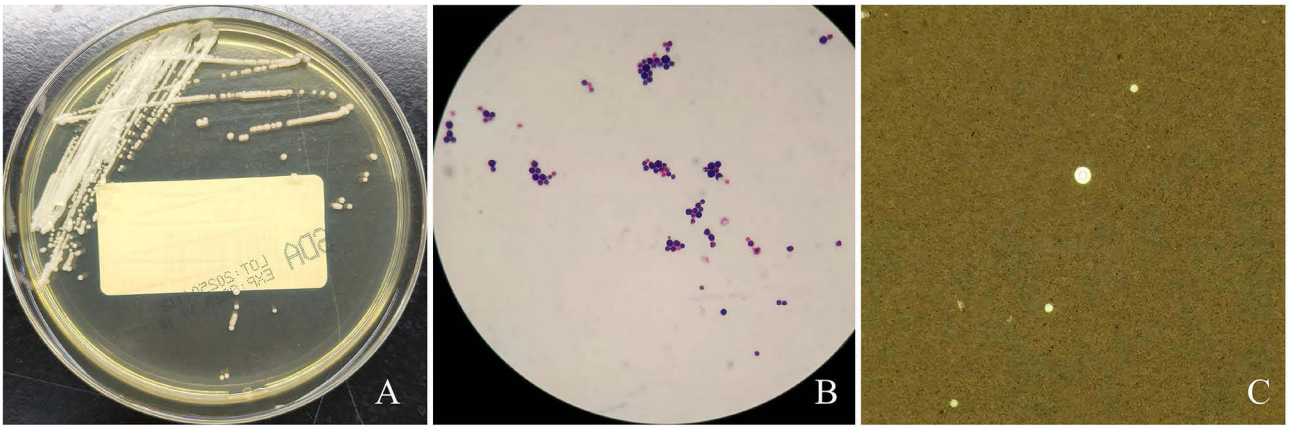


Figure 1 Colonial and microscopic morphology of *C. neoformans* from blood culture. (**A**) *C. neoformans* on Sabouraud Dextrose Agar after 48 hours at 28°C. (**B**) Gram staining of the colony smear. (**C**) India ink staining of the colony, revealing the characteristic capsule of *C. neoformans*.

Treatment

Postoperative anti-infective therapy was administered with meropenem 1.0 g every 8 h combined with omadacycline (loading dose 200 mg once daily, maintenance dose 100 mg once daily). On day 68 of admission, micafungin 150 mg once daily and nebulized amphotericin B 5 mg two to three times daily were added after BALF mNGS detected *C. neoformans* and *Candida auris*. On day 72, as the urinary tract infection worsened, the regimen was changed to amphotericin B 5 mg once daily for bladder irrigation to treat *Candida albicans* based on urine culture results. On day 78, following identification of *C. neoformans* in blood culture, nebulized amphotericin B was replaced with intravenous liposomal amphotericin B 150 mg once daily to treat the bloodstream infection. On day 81, after *C. neoformans* was isolated from urine and pleural fluid cultures, flucytosine 0.5 g every 6 h was added to the anti-infective regimen. Although infection-related markers improved after treatment, subsequent multiple organ failure occurred, and the patient's family requested discharge after 98 days of hospitalization. Key diagnosis and therapeutic milestones are summarized in Table 2.

Discussion

With the increasing global use of organ transplantation and immunosuppressive agents, a rising epidemiological burden of non-HIV-associated cryptococcal infection has been observed.[11] Significant geographical variation has been reported in the distribution of this infection. In the United States, Australia, New Zealand, and most regions in China, more than 90% of cryptococcal infections occur in individuals without HIV infection.[12–14] In contrast, in regions with hot and

Table 1 Antifungal Susceptibility Results of *C. neoformans* Isolated from Blood, Urine and Pleural Effusion Cultures

Antifungal Agent	MIC from Blood (μg/mL)	MIC from Urine (μg/mL)	MIC from Pleural Fluid (μg/mL)
Anidulafungin	8	4	4
Caspofungin	8	8	8
Micafungin	16	16	16
Posaconazole	0.25	0.06	0.06
Voriconazole	0.03	0.03	0.03
Itraconazole	0.25	0.12	0.12
Fluconazole	1	1	1
Amphotericin B	1	2	2
Flucytosine	4	8	8

Abbreviation: MIC, minimum inhibitory concentration.

https://doi.org/10.2147/IDR.S597466

Table 2 Clinical Timeline of Diagnosis and Therapeutic Interventions

Treatment	Day 65	Day 66	Day 68	Day 71	Day 72	Day 76	Day 78	Day 79	Day 81	Day 84	Day 85	Day 88
Meropenem 1.0 g Q8H	+	+	+	+	Stop							
Omadacycline 100 mg QD	+	+	+	+	+	+	+	+	+	+	Stop	
Amphotericin B 5 mg bid-tid Nebulization			+	+	+	+	Stop					
Micafungin 150 mg QD			+	+	+	+	+	+	Stop			
Ceftazidime/Avibactam 2.5 g Q8H					+	+	+	+	+	+	+	+
Amphotericin B 5 mg QD Bladder Irrigation					+	+	+	+	+	+	+	+
Liposomal Amphotericin B 150 mg QD							+	+	+	+	+	+
Flucytosine 0.5 g Q6H									+	+	+	+

Notes: "+" indicates the drug was administered on that day; "stop" indicates the drug was discontinued on that day and not resumed thereafter. Blank cells indicate the drug was not used during that period.
Abbreviations: QD, once daily; Q8H, every 8 hours; bid, twice daily; tid, three times daily.

humid climates or high HIV prevalence, such as South India and Brazil, although HIV-infected individuals still constitute the majority of cases, an increasing trend in non-HIV infections has also been documented.[15,16] Notably, the proportion of hospitalizations due to non-HIV-associated cryptococcal meningitis has increased from 16% in 1997 to 29% in recent years,[17] and the 90-day mortality rate in these patients has reached 41.7%, exceeding the 8.3% rate reported among HIV-associated cases.[6] These findings highlight the substantial challenges in the early diagnosis and clinical management of non-HIV-associated cryptococcosis. Because of their insidious onset and the lack of specific clinical manifestations, delays in diagnosis and treatment are common, underscoring the urgent need for strengthening early recognition and systematic management.

In this context, the present case provides a critical illustration of the diagnostic value and interpretative challenges of mNGS in disseminated cryptococcosis with low fungal burden. The patient's BALF culture was negative, but mNGS detected *C. neoformans* with only 8 sequence reads. Against the clinical backdrop of post-cardiac surgery critical illness and non-response to antibacterial therapy, this low-read signal should have served as an early warning. However, the clinical team hesitated to initiate systemic antifungal therapy due to concerns over false positivity, reflecting a common dilemma in practice where low sequence reads are easily misinterpreted as background noise or contamination. According to the expert consensus of the Chinese Thoracic Society, when a small number of *C. neoformans* reads are detected by BALF mNGS, other rapid tests (such as serum/CSF CrAg and multi-site culture) should be performed concurrently to avoid delay in early anticryptococcal therapy and reduce the risk of disease progression and multiorgan failure.[18] Meanwhile, it is important to recognize that the thick polysaccharide capsule of *Cryptococcus* may reduce nucleic acid extraction efficiency, leading to an underestimation of read counts.[10] Therefore, low-read results should not be readily dismissed but should be integrated with multiple diagnostic modalities for comprehensive interpretation.

To achieve the above integrated interpretation, it is necessary to clarify the advantages and limitations of each diagnostic modality. Traditional diagnostic methods for cryptococcosis, such as India ink staining and fungal culture, are frequently characterized by low sensitivity and specificity because of insufficient fungal load in clinical samples.[19] In contrast, mNGS detects pathogens even in culture-negative samples or after antibiotic administration, and may identify co-infections.[20] The cryptococcal capsular polysaccharide antigen (CrAg) test offers high sensitivity and specificity, reported as 99.7% and 94.1% in serum, and 98.8% and 99.3% in CSF, respectively.[21] CrAg positivity may precede clinical symptom onset by approximately 22 days, considerably earlier than culture or India ink staining.[22] The CrAg lateral flow immunoassay is simple, rapid, and applicable to various sample types, including serum, CSF, and Surine. It is also recommended by the WHO for screening HIV-associated cryptococcal infection.[23] In this patient, CSF smear, culture, and mNGS were negative, while the CrAg test was positive. Combined with imaging and clinical presentation, these results supported the clinical diagnosis of cryptococcal meningitis. For critically ill patients with suspected disseminated infection, especially when atypical sites are involved, concurrent use of BALF mNGS for rapid screening and multisite CrAg testing is recommended, together with proactive multisite cultures to obtain antifungal susceptibility results.

A USA multicenter study reported that non-HIV cryptococcosis most commonly manifests as pulmonary (64.1%) or CNS involvement (49.0%), while infections in the skin, bloodstream, and joints each account for less than 5% of cases.[24] In contrast, urinary tract infections are relatively rare, with an estimated incidence of approximately 0.56 per 10,000 hospital discharges.[25] The detection of *C. neoformans* in urine often suggests disseminated infection, although isolated urinary tract infections with negative blood cultures and serum cryptococcal antigen testing have also been described.[26] *C. neoformans* infects the urinary tract through two mechanisms. One is hematogenous dissemination; after initial pulmonary invasion through the respiratory route, the fungus enters the bloodstream, reaching the kidneys and causing infection, particularly in immunocompromised patients. The other mechanism is ascending infection, facilitated by invasive urinary procedures such as catheterization or ureteral stent placement or by the disruption of the mucosal barrier.[25] Diagnostic clues include proteinuria or visualization of budding yeast cells on urinalysis.[25] Therefore, heightened vigilance for abnormal yeast morphology in urine is required to prevent missed diagnoses.

Pleural infection represents another rare site for cryptococcal infection. In this case, the initial pulmonary infiltrates accompanied by pleural effusion were nonspecific and susceptible to misdiagnosis. Pleural involvement primarily occurs through hematogenous dissemination, with Cryptococci disseminating from the lung to the pleura through the bloodstream, or by direct extension from adjacent lung parenchymal lesions that breach the pleural surface, inducing inflammatory exudate.[27] Diagnostic confirmation is challenging, as traditional India ink staining of pleural fluid has a very low positivity rate and pleural fluid culture is positive in only 42% of cases because of slow growth and low organism burden.[27] Furthermore, both tuberculous and cryptococcal pleurisy may present with lymphocyte-predominant pleural effusion and elevated adenosine deaminase levels, requiring careful differentiation.[28,29] A previous study demonstrated that CrAg testing in pleural fluid may achieve sensitivity up to 89%, making it an important tool for early and rapid diagnosis.[30] Combined assessment with serum CrAg further assists in evaluating disseminated infection.

Additionally, a noteworthy finding was the simultaneous detection of *C. auris* sequences in the same BALF sample by mNGS (2,649 reads). Although the WHO has classified *C. auris* as a "critical" priority fungal pathogen,[31] we interpreted this as respiratory colonization rather than invasive co-infection based on three key observations: (1) routine fungal culture of the same BALF sample was negative; (2) throughout hospitalization, cultures from sterile sites including blood, urine, pleural fluid, and cerebrospinal fluid all showed no growth of *C. auris*; and (3) the patient showed clinical and inflammatory improvement following anti-Cryptococcus therapy. This case highlights the sensitivity of mNGS in detecting multidrug-resistant organisms. Second, it cautions that a high read count for a clinically sensitive pathogen like *C. auris* does not automatically signify invasive disease. Thus, mNGS results must be integrated with conventional cultures, clinical presentation, and treatment response to avoid overdiagnosis and unnecessary broad-spectrum antifungal therapy.

This report has several limitations. As a single case, the findings may not be generalizable. The interpretation of mNGS results, particularly low sequence read counts, remains challenging and may be influenced by technical factors. In addition, negative CSF culture and mNGS despite clinical suspicion of CNS involvement may reflect a low fungal burden. These limitations highlight the need for multimodal diagnostic integration and clinical correlation.

Conclusion

This case highlights that disseminated cryptococcosis in non-HIV patients may present with atypical organ involvement and low fungal burden, leading to diagnostic delay. The detection of *C. neoformans* by mNGS with low sequence reads provided an early diagnostic clue, while subsequent culture and cryptococcal antigen testing were essential for confirmation, illustrating the complementary roles of these modalities. The involvement of uncommon sites, such as the urinary tract and pleura, should raise suspicion for disseminated disease. Early implementation of multimodal diagnostic strategies and timely initiation of antifungal therapy are critical to improving outcomes in this high-risk population.

Data Sharing Statement

Data will be available upon reasonable request from the corresponding author (Pingjuan, Liu, Email: liupj8@mail.sysu.edu.cn).

Ethics Approval and Consent to Participate

This study complies with the Declaration of Helsinki. The study protocol and the publication of case details were approved by the First Affiliated Hospital of Sun Yat-sen University (Approved No. of ethics committee: 2025503). The patient provided written informed consent for the publication of this case report, in accordance with the journal's consent policy.

Funding

There is no funding to report.

Disclosure

The authors report no conflicts of interest in this work.

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